

Research Article

Multiscale Features Integrated Model for Generalizable Deepfake Detection

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Within the domain of Artificial Intelligence Generated Content (AIGC), technological strides in image generation have been marked, resulting in the proliferation of deepfake images that pose substantial security threats. The current landscape of deepfake detection technologies is marred by limited generalization across diverse generative models and a subpar detection rate for images generated through diffusion processes. In response to these challenges, this paper introduces a novel detection model designed for high generalizability, leveraging multiscale frequency and spatial domain features. Our model harnesses an array of specialized filters to extract frequency-domain characteristics, which are then integrated with spatial-domain features captured by a Feature Pyramid Network (FPN). The integration of the Attentional Feature Fusion (AFF) mechanism within the feature fusion module allows for the optimal utilization of the extracted features, thereby enhancing detection capabilities. We curated an extensive dataset encompassing deepfake images from a variety of GANs and diffusion models for rigorous evaluation. The experimental findings reveal that our proposed model achieves superior accuracy and generalization compared to existing baseline models when confronted with deepfake images from multiple generative sources. Notably, in cross-model detection scenarios, our model outperforms the next best model by a significant margin of 29.1% for diffusion-generated images and 15.1% for GAN-generated images. This accomplishment presents a viable solution to the pressing issues of generalization and adaptability in the field of deepfake detection.

Keywords: Artificial Intelligence Generated Content; deepfake detection; diffusion model; generative adversarial networks

1. Introduction

As advancements in Artificial Intelligence Generated Content (AIGC) technologies continue to accelerate, a multitude of detection methods have been devised to authenticate synthetic media [1–3], tackling challenges such as the spread of misinformation, infringements on personal privacy, and instances of online financial fraud. Notably, deepfake images emerge as a focal point within this domain, owing to their propensity for rapid dissemination and their ubiquity on social media platforms. Therefore, this research underscores

the importance of developing robust detection techniques specifically tailored to deepfake images.

Currently, the creation of deepfake images is primarily facilitated by two leading generative models: generative adversarial networks (GANs) [4] and diffusion models (DMs) [5]. Detection models for GAN-generated images typically enhance accuracy (ACC) by refining feature extraction, optimizing network architectures, and incorporating data augmentation strategies. Feature-based approaches primarily target the identification of spatial and frequency domain inconsistencies. For instance, McCloskey et al. [6] leveraged

red-green bivariate histograms and abnormal pixel exposure ratios for detection, while Agarwal et al. [7] exploited high-frequency artifacts stemming from GANs' upsampling processes. Chen et al. [8] integrated both global and local image features, utilizing metric learning to enhance the overall detection performance of model. In terms of network architectures, Convolutional Neural Networks (CNNs) remain a predominant choice for deepfake detection tasks due to their capacity to extract semantic, color, and texture information [9], as seen in Fu et al. [10] dual-channel CNN architecture, which is capable of concurrently processing both high- and low-frequency image components. Additionally, data augmentation plays a crucial role in boosting robustness and generalization. Wang et al. [11] demonstrated that combining preprocessing, postprocessing, and data augmentation allows a CNN trained on a single GAN dataset to generalize across multiple GAN models effectively.

For the detection of DM generated images, a straightforward approach involves adapting existing GAN-based detectors. However, Corvi et al. [12] demonstrated that even advanced GAN detection algorithms exhibit suboptimal performance when tasked with identifying DM-generated images. In a manner similar to GAN-generated image detection, several studies leveraged spatial and frequency domain features for detecting DM images. For instance, Nguyen et al. [13] employed gradient-based features, while Bammey et al. [14] used cross-differential filters to highlight frequency artifacts. Additionally, Farid et al. [15, 16] highlighted inconsistencies in lighting, geometric structures, shadows, and reflections present in DM-generated images. Ojha et al. [17] proposed an alternative method by mapping images into CLIP's feature space and classifying them using cosine similarity with reference real or fake images. WANG et al. [18] introduced a method using the difference between reconstructed images and those under examination as features, achieving high ACC in the detection of diffusion-generated images.

Despite these developments, deepfake detection methodologies encounter numerous challenges. For GAN-generated images, existing detection models tend to perform inadequately when applied to multisource datasets due to their constrained feature extraction dimensions. While data augmentation and dataset expansion have been employed to enhance generalization [11], they often yield limited effectiveness due to their predictability. Furthermore, the robustness of GAN-based detectors is often compromised in the face of adversarial perturbations, as evidenced by the localized attacks conducted by Zhang et al. [19], which revealed vulnerabilities in a range of detection models. As for diffusion-generated images, effective detection techniques remain underexplored, and those available tend to be overly complex, lacking comprehensive image feature analysis, thus hindering their practical applicability.

In real-world scenarios, the generative model responsible for creating deepfake images is often unknown, emphasizing the necessity for detection models with strong cross-model generalization capabilities. However, traditional detection models struggle to handle such uncertainty, for they often fail to adequately leverage the rich and complex features inherent in deepfake images, or rely solely on single-dimensional

feature learning, which limits their ability to capture the shared characteristics across different generative models. Ricker et al. [20] demonstrated that GAN detectors often fail to detect images generated by DMs, and retraining these detectors on diffusion-generated datasets leads to only marginal improvements. This lack of adaptability exposes a gap in current detection technologies, which struggle to handle the variety of deepfake generation techniques and their artifacts. To address the growing concerns surrounding synthetic media manipulation, there is an urgent need for robust and adaptable models capable of detecting deepfakes across diverse platforms and applications, regardless of the generative model or source datasets employed.

To address these limitations, this paper introduces a generalizable deepfake detection model that leverages both spatial and frequency domain features commonly exhibited by deepfake images. The proposed model utilizes a series of filters to extract multiscale frequency-domain features, and a Feature Pyramid Network (FPN) [21] to capture multiscale spatial-domain features. These features are subsequently fused and processed through a ResNet50 backbone for classification. To fully exploit the extracted features, multiple fusion strategies incorporating the Attentional Feature Fusion (AFF) [16] mechanism are devised, further enhancing the model's performance. The primary contributions of this work are as follows: (1) We develop a model-independent detection method that effectively identifies forged images from various generative models and datasets, addressing the black-box detection challenge where the generative model is unknown. (2) By extracting and fusing multiscale features from both spatial and frequency domains, we capture shared characteristics across different generative models, overcoming the limitations of single-feature modeling. (3) Experimental results demonstrate that our model significantly outperforms existing detectors on both GAN-generated and diffusion-generated images, as well as on images generated by advanced models, exhibiting superior generalization and detection ACC, thus validating the effectiveness of our approach.

2. The Proposed Method

2.1. Overall Structure. The overall structure of the proposed model is illustrated in Figure 1, comprising two primary components: a feature extraction module and a feature fusion module. In the feature extraction module, the images are initially processed through an image pyramid to create multiscale representations, which are subsequently passed through a series of filters to extract multiscale frequency-domain features. Concurrently, the original input images are fed into a FPN, facilitating the extraction of multiscale spatial-domain features. In the subsequent feature fusion module, the extracted features from both domains are effectively integrated through the AFF mechanism, ensuring an effective combination of multiscale spatial and frequency information. The fused features are then processed by a ResNet50 backbone network, which ultimately classifies the image as either real or fake. The detailed architecture and functionality of the model will be elaborated in the following sections.

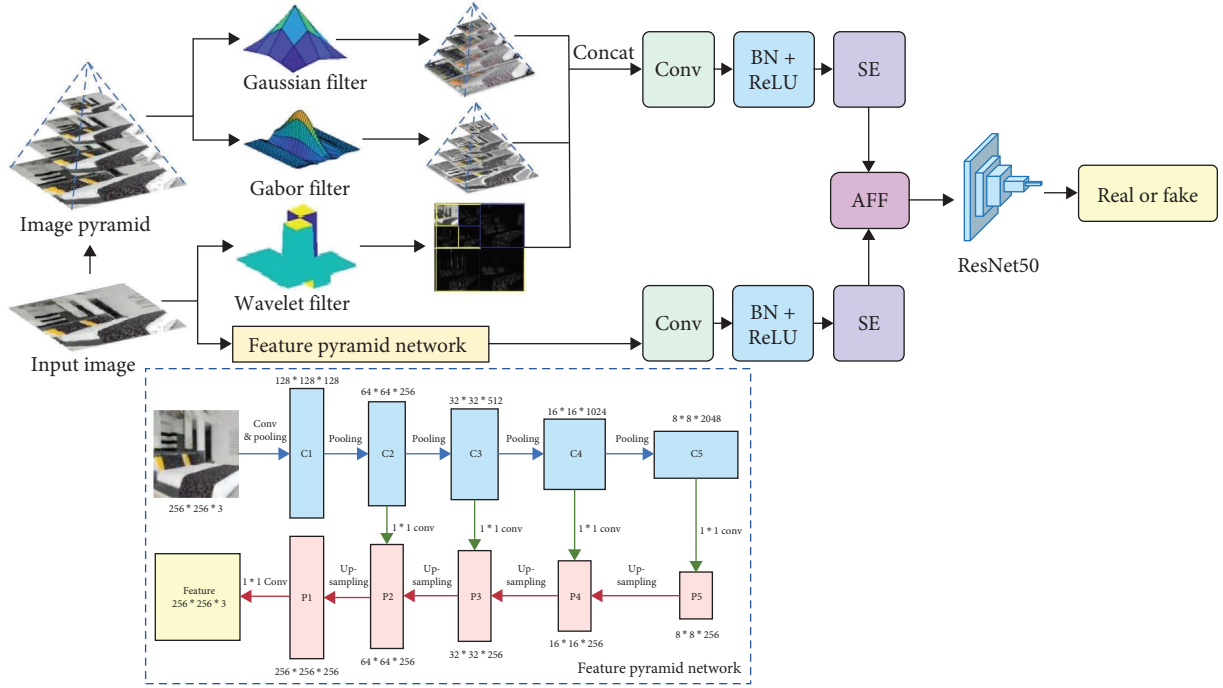


FIGURE 1: Overall structure of the multiscale features integrated model for deepfake detection.

2.2. Frequency-Domain Feature Extraction. The generation of deepfake images involves sophisticated synthesis techniques, such as pixel-level manipulation, temporal and spatial consistency adjustments, and fine-tuning of facial features. These manipulations often introduce subtle discrepancies between deepfake and authentic images, which are most effectively observed in the frequency domain. However, these inconsistencies do not exhibit uniform patterns, necessitating the extraction of subtle variations and anomalous features across multiple scales. To address this, we employ a combination of Gaussian, Gabor, and wavelet filters, each selected for its specific advantages in extracting relevant frequency-domain features.

Gaussian filters are employed to capture low-frequency information, which is essential for identifying overall structural elements and large-scale patterns in images. As fundamental techniques for smoothing and noise reduction, they are particularly suitable for extracting the broad, global characteristics of deepfake images, which are often affected by low-frequency distortions. Specifically, a 3×3 Gaussian convolution kernel with a standard deviation of 1 and mean values of one is applied. The two-dimensional Gaussian function is defined as follows:

$$\text{Gauss}(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\left[\frac{(x-\mu_x)^2}{2\sigma_x^2} + \frac{(y-\mu_y)^2}{2\sigma_y^2}\right]\right), \quad (1)$$

where $\text{Gauss}(x, y)$ represents the Gaussian weight at point (x, y) , σ_x, σ_y are the standard deviations, μ_x, μ_y are the mean values, and the normalization factor $1/2\pi\sigma_x\sigma_y$ ensures the sum of kernel equals 1. To capture multiscale features, input

images are downsampled by a factor of 2 using an image pyramid before Gaussian filtering, followed by linear interpolation and pixel-wise summation across scales. This procedure integrates multiscale filtered information while preserving critical edges and details.

Gabor filters, renowned for their ability to detect edges and analyze textures, are employed for their robust capability in directional and scale-selective analysis. These filters are particularly effective in isolating mid-to-high-frequency components and detecting local patterns, which are often indicative of subtle artifacts introduced by deepfake generation techniques. Their ability to capture directional and textural information makes them well-suited for analyzing textured regions in deepfake images. The two-dimensional Gabor function is defined as follows:

$$\text{Gabor}(x, y) = \exp\left(-\frac{x'^2 + y'^2}{2\sigma^2}\right) \cdot \cos\left(2\pi\frac{x'}{\lambda} + \psi\right), \quad (2)$$

where σ is the Gaussian standard deviation, γ is the spatial aspect ratio, λ is the wavelength, and ψ is the phase offset. The rotated coordinates x' and y' are given in equations (3) and (4):

$$x' = x \cos \theta + y \sin \theta, \quad (3)$$

$$y' = -x \sin \theta + y \cos \theta, \quad (4)$$

where θ defines the filter's orientation. In this work, we employ 16 Gabor kernels with orientations ranging from 0° to 360° , each with a kernel size of 31×31 , a standard deviation of 4, a spatial aspect ratio of 0.5, a wavelength of 10,

and a phase offset of 0. These filters process multiscale images generated by the image pyramid, capturing texture and directional features from various orientations.

Wavelet filters are particularly effective in balancing spatial and frequency domain information, enabling a comprehensive multiscale analysis of deepfake images. They allow the separation of high- and low-frequency components, facilitating the detection of both large-scale structural inconsistencies and localized anomalies. Wavelet decomposition provides a powerful tool for extracting multiresolution features, which can be crucial for identifying subtle artifacts that are distributed across different frequency bands. To harness these capabilities, the Daubechies 4-tap (db4) wavelet is selected due to its balance between computational efficiency and its ability to capture detailed image features across scales. The wavelet decomposition and reconstruction processes are represented by equations (5) and (6), respectively:

$$\text{DWT}(I) \longrightarrow (LL, LH, HL, HH), \quad (5)$$

$$\text{high_freq_img} = \sum(|LH| + |HL| + |HH|), \quad (6)$$

where I represents the input image. LL denotes the low-frequency approximation coefficients capturing the overall structure and large-scale features of the image. LH , HL , HH represent the high-frequency detail coefficients in the horizontal, vertical, and diagonal directions, respectively. The high-frequency images are combined to generate a complete high-frequency image using equation (6), which, along with the low-frequency approximation LL , is utilized for detection. Given the wavelet transform's inherent capability to effectively represent multiscale features, the original image is utilized directly as input, bypassing the need for pyramid preprocessing. The wavelet decomposition is conducted over two levels, producing multiscale wavelet-filtered representations that encapsulate both global and localized image characteristics. The db4 wavelet enables the comprehensive utilization of an image's detailed information and noise characteristics, thereby enabling robust analysis of both global and local anomalies in deepfake images.

Collectively, the integration of Gaussian, Gabor, and wavelet filters facilitates the extraction of comprehensive frequency-domain features. Gaussian filters capture global structural patterns by focusing on low-frequency information. Gabor filters extract localized texture and edge information through their directional and scale-selective properties. Wavelet filters provide a multiscale analysis by decomposing the image into various frequency components. Together, these filters enhance the model's ability to identify subtle inconsistencies across scales, leading to more effective deepfake detection.

2.3. Spatial-Domain Feature Extraction. In the previous section, it was illustrated that the selected filters effectively captured multiscale features of frequency domain, including texture and edge information. However, relying exclusively on the frequency-domain features imposes inherent

limitations on the detection model's ability to fully encapsulate the semantic content of images. Generative models often introduce subtle artifacts in both global structures and localized details, necessitating the extraction of complementary spatial-domain features for a more comprehensive analysis. By synthesizing both spatial and frequency-domain characteristics, the model achieves a more holistic representation, significantly improving the detection performance, particularly in complex and heterogeneous scenarios.

In this research, we utilize a FPN [21] architecture, which is constructed on the foundation of a ResNet50 [22] backbone, to effectively extract multiscale spatial-domain features. During the bottom-up pathway of the FPN, corresponding to the forward propagation within ResNet50, four feature maps of distinct resolutions are produced. These feature maps progressively distill high-level semantic information through multiple convolutional layers, while simultaneously preserving essential multiscale structural information. In the top-down pathway, these feature maps are subjected to dimensionality reduction via 1×1 convolutional layers, upsampled using linear interpolation, and then merged layer by layer to integrate semantic information across various scales. The fused feature maps are subsequently refined through additional convolution layers to yield the final multiscale feature representation. The structure of the FPN is shown in Figure 2. By these means, the FPN can effectively extract and synthesize multiscale spatial-domain features, enabling the model to effectively perceive both overarching semantic structures and intricate local details, thereby substantially improving detection ACC and generalization across diverse scenarios.

2.4. Feature Fusion Module. In the preceding sections, we selected a variety of complementary frequency-domain and spatial-domain features to enhance the efficacy of deepfake detection. By integrating these features, a more holistic and robust image representation can be constructed. These features are processed through a Batch Normalization (BN) layer, followed by ReLU activation, and then passed through a Squeeze-and-Excitation (SE) attention layer before being input into a ResNet50 network for final classification.

To further optimize the utilization of extracted features, we incorporate the AFF [23] mechanism within the feature fusion module. AFF is an adaptive fusion technique designed to dynamically combine input features in a scale-aware and content-adaptive manner. This mechanism is particularly effective in addressing the challenge of merging features with disparate scales, resolutions, or semantic meanings, which is a common issue when integrating spatial and frequency-domain features.

The AFF framework consists of two key components: feature aggregation and attention-based weighting. The feature aggregation process first combines the input features through element-wise addition to produce a unified representation. To address the limitations of raw aggregation, which fails to account for the varying importance of features across different scales and contexts, attention-based

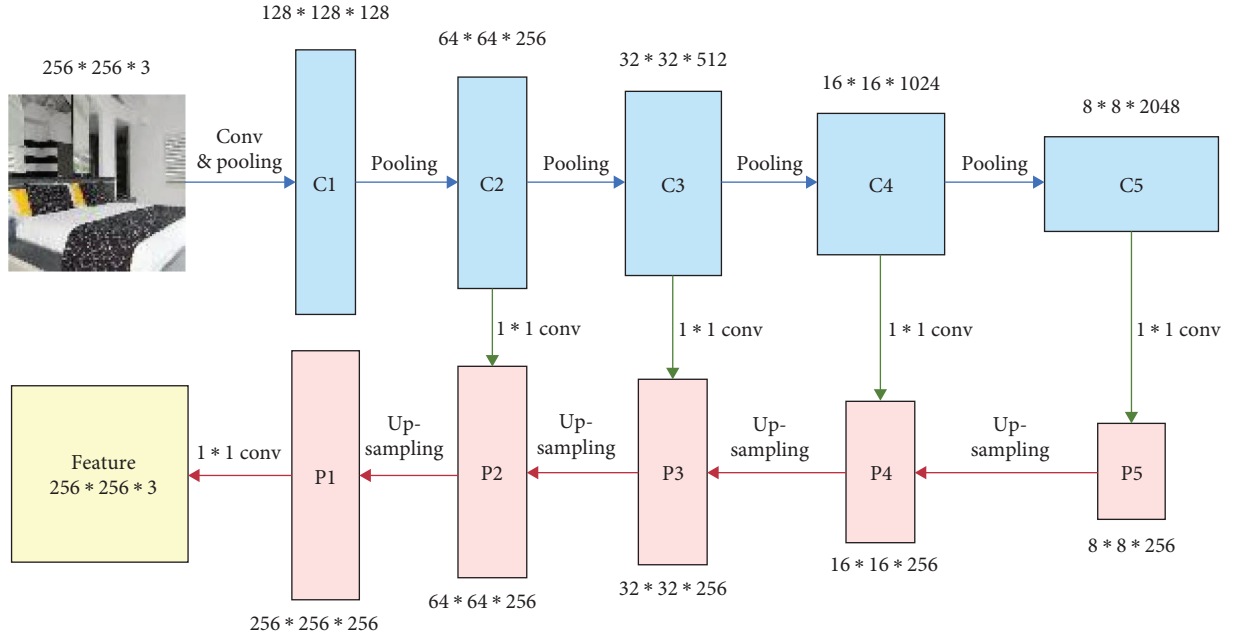


FIGURE 2: Structure of Feature Pyramid Network.

weighting is employed. The Multiscale Channel Attention Module (MS-CAM) calculates dynamic attention weights for each feature, facilitating the selective enhancement of more informative features while suppressing less relevant ones. This enables the model to prioritize features that are most indicative of subtle inconsistencies in deepfake images, thereby refining the fusion process and improving detection ACC. The structures of AFF and MS-CAM are depicted in Figures 3 and 4, respectively.

The specific process of the AFF strategy is shown in equation (7). Given two input features X and Y , the initial fusion is performed through element-wise addition, producing a preliminary combined feature $X \oplus Y$. This is followed by the computation of a weight $M(X \oplus Y)$ via MS-CAM, which modulates the relative significance of X and Y . The final fused feature Z is derived by applying a weighted sum, where $M(X \oplus Y)$ is assigned to feature X and $1 - M(X \oplus Y)$ to feature Y . Although this approach introduces a slight increase in computational cost, the additional FLOPs are minimal, making it a favorable trade-off in light of the significant improvements in detection performance.

$$Z = M(X \oplus Y) \otimes X + (1 - M(X \oplus Y)) \otimes Y. \quad (7)$$

3. Experiment

3.1. Datasets and Experiment Setting. In this study, we comprehensively evaluate the performance of model by constructing a diverse dataset that includes both diffusion-generated and GAN-generated images, as well as more recent deepfake images from advanced models. The dataset is carefully curated to include images from a variety of generative models, ensuring a broad representation of different architectures and data sources. As outlined in Table 1, the

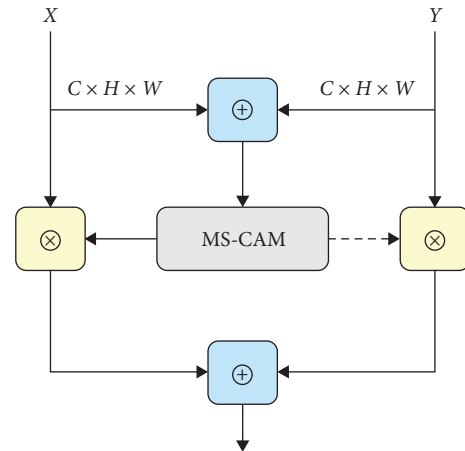


FIGURE 3: Structure of Attentional Feature Fusion Module.

training set includes images generated by ProGAN [26], a GAN-based model, and ADM [24], a DM. The testing set is designed to provide a comprehensive evaluation, including models like iDDPM [27], PNDM [28], and ADM [24] trained on datasets such as LSUN_bedroom, ImageNet, CelebA, as well as more recent models like Midjourney [34] and DALLE2 [35]. Notably, there are no overlapping samples between the training and testing sets, which ensures an unbiased evaluation of the model's generalization capabilities and mitigates the risk of overfitting. These sources are selected to ensure both model diversity and real-world applicability, covering a wide range of generative techniques.

During the training phase, the dataset is split into training and validation subsets in an 80–20 ratio to facilitate cross-validation and ensure a fair evaluation of the model's performance. To enhance the model's robustness and generalization, data augmentation techniques such as random

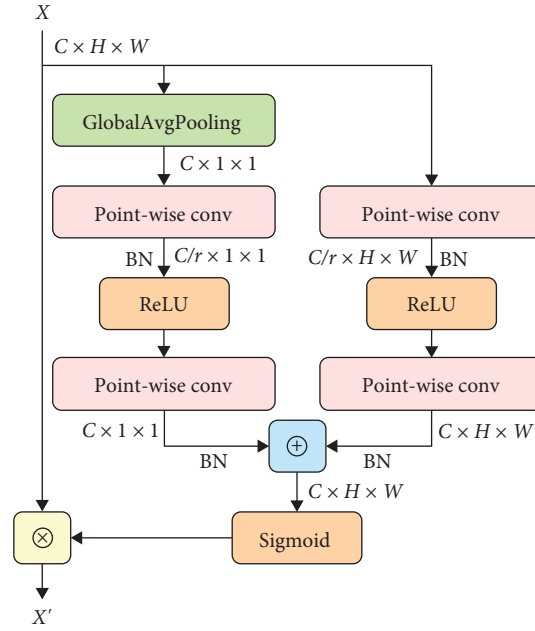


FIGURE 4: Structure of Multiscale Channel Attention Module.

TABLE 1: Deepfake image dataset containing various types of generated images.

Data type	Model type	Generative model	Source dataset	Volume (real/fake)
Training	DM	ADM [24]	LSUN_B [25]	20k/20k
	GAN	ProGAN [26]	ProGAN	10k/10k
Testing	DM	iDDPM [27]	LSUN_B	2k/2k
		PNDM [28]	LSUN_B	2k/2k
		ADM [24]	LSUN_B	2k/2k
		ADM [24]	ImageNet [29]	2k/2k
		SDv2 [30]	CelebA	2k/2k
	GAN	bigGAN [31]	ImageNet	2k/2k
		cycleGAN [32]	cycleGAN	2k/2k
		starGAN [33]	CelebA	2k/2k
		Midjourney [34]	CelebA	2k/2k
		DALLE2 [35]	CelebA	2k/2k

rotation (with a ± 15 -degree range) and random horizontal flipping are applied, and all images are uniformly center-cropped to 256×256 pixels.

The model is implemented using the PyTorch framework, version 2.0.0. Its performance is evaluated using two metrics: ACC and F1-score. ACC measures overall detection ACC, while the F1-score accounts for the balance between precision and recall, which is particularly crucial in scenarios with potential class imbalance. For the training process, binary cross-entropy is chosen as the loss function, as the task involves binary classification. The model is trained using the Adam optimizer with an initial learning rate of 0.0001 and a batch size of 32 for 50 epochs. A learning rate decay strategy is applied, where the learning rate is reduced to 10% of its current value if validation metrics do not improve over 8 consecutive epochs. Training is conducted on an NVIDIA V100 GPU to leverage its computational capabilities. The environment setup includes CUDA 11.7 and cuDNN 8 to ensure efficient utilization of GPU resources.

3.2. Ablation Experiment of Three Filters. In this experiment, we aim to evaluate the independent and combined contributions of three frequency-domain filters to the overall performance of our model. The models under investigation are as follows: (1) Origin: Directly feeding the original input images into the ResNet50 network without applying any frequency-domain filtering; (2) Gauss: Processing the images with a Gaussian filter before feeding them into the ResNet50 backbone; (3) Gabor: Processing the images with a Gabor filter before feeding them into the ResNet50 backbone; (4) Wave: Processing the images with a Wavelet filter prior to feeding them into the ResNet50 backbone; (5) Gauss_Gabor: Concatenating the feature maps from the Gaussian and Gabor filters before feeding them into the ResNet50 backbone; (6) Gauss_Wave: Concatenating the feature maps from the Gaussian and Wavelet filters before feeding them into the ResNet50 backbone; (7) Gabor_Wave: Concatenating the feature maps from the Gabor and Wavelet filters before feeding them into the ResNet50

backbone; (8) **Combined**: Integrating the outputs of the Gaussian, Gabor, and Wavelet filters by concatenating their features, followed by processing them through convolutional layers before feeding into the ResNet50 backbone.

For this study, we use the ADM and ProGAN datasets described in Table 1 for training, while the testing set comprises images from all datasets enumerated in Table 1. Specifically, 500 real images and 500 generated images are randomly selected from each of the eight testing sets, resulting in a comprehensive mixed testing set to evaluate model performance comprehensively. The evaluation metrics include detection accuracy and F1-scores, which are presented in Table 2 for each filtering strategy.

The experimental results indicate that all three individual filtering strategies (Gauss, Gabor, Wave) outperform the origin model, confirming the effectiveness of each filter. Furthermore, the results of the two-filter combinations (Gauss_Gabor, Gauss_Wave, Gabor_Wave) using concatenation demonstrate varying improvements in detection performance, with the combined model, which integrates all three filters by concatenating their feature maps, achieving the highest performance. This suggests that while each filter contributes positively, the combination of all three filters leverages the complementary strengths of the three filters, with Gaussian filtering enhancing local smoothing, Gabor kernels capturing texture orientation, and wavelets highlighting multiscale features, leading to notable improvements in both accuracy and robustness. These findings highlight the value of integrating diverse frequency-domain features to enhance model performance in complex detection tasks.

3.3. Ablation Experiment of Feature Fusion Strategy. In this experiment, we evaluate the performance of three distinct feature fusion strategies. The strategies under investigation are as follows: (1) Concat: Concatenating all the extracted features; (2) AFF1: Applying AFF strategy to fuse the multiscale frequency-domain features extracted from the three filters, followed by concatenation with spatial-domain features derived from FPN; (3) AFF2: Initially concatenating the frequency-domain features, then fusing them with the spatial-domain features using AFF. The detailed process of each fusion strategy is illustrated in Figure 5. The same training and testing sets used in Section 3.2 are employed in this experiment to maintain consistency. The model performance, in terms of detection ACC and F1-scores for each fusion strategy, is presented in Table 3.

The results demonstrate that the model employing AFF2 fusion strategy achieves superior ACC and F1-score compared to other models, which can be largely attributed to the complementary nature of multiscale frequency-domain features. The application of AFF solely to frequency-domain features (AFF1) yields limited improvement. In contrast, when AFF is employed to fuse both frequency- and spatial-domain features (AFF2), it facilitates a more effective integration of cross-scale and cross-modal information, significantly enhancing overall model performance. Besides, the Concat model shows slightly inferior performance in

TABLE 2: Accuracy and F1-scores of models using different filtering strategies.

Training set Metrics	ProGAN		ADM	
	ACC	F1	ACC	F1
Origin	0.501	0.003	0.509	0.040
Gauss	0.752	0.741	0.806	0.836
Gabor	0.717	0.712	0.782	0.761
Wave	0.776	0.756	0.807	0.837
Gauss_Gabor	0.762	0.754	0.823	0.839
Gauss_Wave	0.789	0.778	0.838	0.854
Gabor_Wave	0.782	0.763	0.831	0.847
Combined	0.803	0.792	0.853	0.865

Note: The highest values are highlighted in bold to emphasize the effectiveness of the adopted modules and the superiority of our proposed model.

comparison with the combined model in Section 3.2, likely due to its lack of effective mechanisms for integrating complementary relationships between features. These findings highlight the efficacy of AFF mechanism in deepfake detection, as it maximizes the utility of diverse feature representations while reducing redundancy, thus improving both detection accuracy and generalization.

3.4. Ablation Experiment of FPN. Although the contribution of the FPN module has already been reflected in the results of the previous two ablation experiments, we present this dedicated comparison to ensure the completeness of our ablation study. The models under investigation are as follows: (1) Without FPN: the combined model from Section 3.2. (2) With FPN: the AFF2 model from Section 3.3, which is also the **proposed** model.

The results, as summarized in Table 4, indicate that the model **with FPN** outperforms the model **without FPN** in both detection ACC and F1-score. Specifically, the incorporation of the FPN allows the model to more effectively capture spatial information at multiple scales, which complements the frequency-domain features and results in enhanced detection performance. These findings highlight the value of integrating spatial-domain features via the FPN, demonstrating that the addition of the FPN module significantly contributes to the model's capability to generalize across different deepfake sources, thereby improving both ACC and robustness.

3.5. Comparison Experiment of Detection ACC. In this experiment, we evaluate the performance of our proposed model in comparison to existing models recognized for their strong generalization capabilities and high detection ACC. Given the superior performance of the AFF2 fusion strategy observed in the ablation studies, we select the model employing this strategy for further comparison. To assess the model's effectiveness in detecting both GAN and diffusion-generated images, we utilize two baseline models: WangCNN [11] and DIRE [18]. WangCNN functions as a general detector for GAN-generated images, demonstrating strong generalization across various GAN datasets using preprocessing, postprocessing, and data augmentation

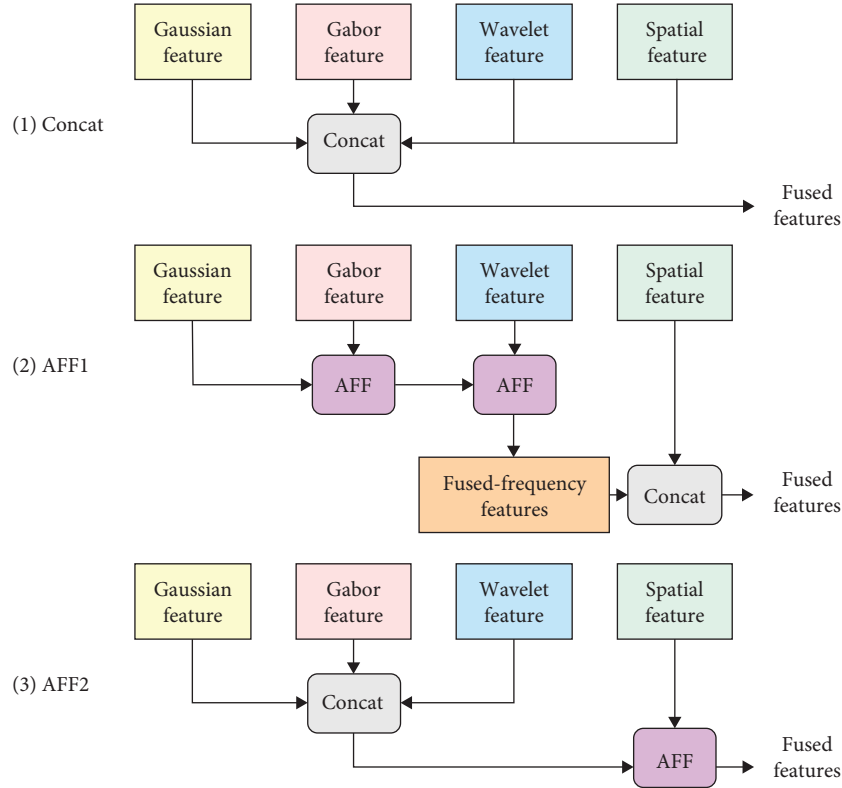


FIGURE 5: Diagram of three feature fusion strategies.

TABLE 3: Accuracy and F1-scores of models under different fusion strategies.

Training set	Metrics	Concat	AFF1	AFF2 (proposed)
ProGAN	ACC	0.799	0.8	0.815
	F1	0.779	0.777	0.795
ADM	ACC	0.842	0.871	0.881
	F1	0.824	0.874	0.891

Note: The highest values are highlighted in bold to emphasize the effectiveness of the adopted modules and the superiority of our proposed model.

TABLE 4: Accuracy and F1-scores of models with and without FPN.

Training set	Metrics	Without FPN	With FPN (proposed)
ProGAN	ACC	0.803	0.815
	F1	0.792	0.795
ADM	ACC	0.853	0.881
	F1	0.865	0.891

Note: The highest values are highlighted in bold to emphasize the effectiveness of the adopted modules and the superiority of our proposed model.

techniques. The DIRE model, on the other hand, is specifically designed for detecting diffusion-generated images through the discrepancies between the input image and its reconstructed counterpart.

Using the ProGAN dataset for training, we compare the performance of our model with the baseline models across images generated by various generative models and datasets, as summarized in Table 5. The two variations of WangCNN,

TABLE 5: Comparison of detection accuracy of different models trained on the ProGAN dataset.

	WangCNN0.1	WangCNN0.5	DIRE	Proposed
ADM_lsun	0.528	0.513	0.521	0.798
iDDPM	0.575	0.536	0.522	0.762
PNDM	0.61	0.559	0.543	0.738
ADM_imag	0.573	0.556	0.679	0.87
SD	0.416	0.54	0.84	0.837
DM_mix	0.54	0.541	0.621	0.802
bigGAN	0.645	0.599	0.662	0.744
cycleGAN	0.817	0.833	0.726	0.773
starGAN	0.828	0.793	0.858	0.974
GAN_mix	0.763	0.742	0.749	0.836
Midjourney	0.553	0.523	0.623	0.863
DALLE2	0.470	0.646	0.806	0.839
Total_mix	0.624	0.616	0.669	0.815

Note: The highest values are highlighted in bold to emphasize the effectiveness of the adopted modules and the superiority of our proposed model.

differentiated by their use of data augmentation techniques, are referred to as WangCNN0.1 and WangCNN0.5. Since the testing set includes images generated by ADM based on two different datasets, they are distinguished as ADM_lsun and ADM_imag in the table. Additionally, DM_mix, GAN_mix, and Total_mix refer to the mixed datasets consisting of diffusion-generated, GAN-generated, and both types of images, respectively. Specifically, DM_mix contains images from five DM testing sets, GAN_mix from three GAN testing sets, and Total_mix from all eight testing sets.

For each mixed dataset, 500 real images and 500 generated images are randomly selected from all constituent datasets.

The experimental results demonstrate that our model, trained on the ProGAN dataset, not only performs well on GAN-generated images but exhibits a particularly notable advantage in detecting diffusion-generated images. On the three mixed datasets, the detection ACC of our model for GAN-generated images (GAN_mix) is 9.6% higher than that of the second-best model WangCNN0.1. Additionally, the most significant performance gain is observed for diffusion-generated images (DM_mix), where our model surpasses the second-best model DIRE by an impressive 29.1%. Overall, our model achieves a 21.8% higher detection ACC than DIRE on the combined testing set (Total_mix), further underscoring its robustness in handling both types of images, with a clear edge in diffusion-generated image detection.

When trained on the ADM dataset, the results in Table 6 reveal that our model consistently delivers superior detection performance for both diffusion- and GAN-generated images. Notably, the model excels in identifying cross-model GAN-generated images, achieving the highest detection ACC across all GAN testing sets, with a 15.1% improvement on the GAN_mix testing set over the second-best model, DIRE.

Furthermore, when tested on the Midjourney and DALLE2 datasets, our model achieves the highest detection ACC compared to other baselines, regardless of the training dataset used. This further reinforces its generalization capabilities across both traditional and newer generative models. The results of the two comparison experiments demonstrate that our model not only maintains high detection ACC for images generated by models of the same type, but also excels in detecting cross-model generated images, highlighting its strong generalization ability across diverse deepfake generation techniques.

3.6. Analysis of Performance Variations. Although the model demonstrates strong performance across most datasets, notable variations in performance are observed across different testing sets. Using the ADM training dataset as a case study, we explore these discrepancies in greater detail.

As shown in Table 6, when tested on the images generated by ADM using the ImageNet dataset (ADM_imag dataset), the model's ACC notably declines to 0.77, compared to 0.985 on the ADM_lsun dataset. This suggests that the higher variability and naturalistic features of ImageNet images may present more challenges for the model, which was primarily trained on the LSUN_bedroom dataset, highlighting a potential gap in the model's ability to generalize to such varied deepfake characteristics.

Additionally, the model's performance on bigGAN (0.715) and cycleGAN (0.711) is lower compared to its performance on starGAN images (0.965). This may be due to the fact that starGAN images are potentially more similar to the images in training set or to diffusion-generated images in general. The more consistent and structured features of starGAN images could align more closely with the model's learned characteristics, which were optimized for detecting patterns found in diffusion-based deepfakes. As a result, this

TABLE 6: Comparison of detection accuracy of different models trained on the ADM dataset.

	WangCNN0.1	WangCNN0.5	DIRE	Proposed
ADM_lsun	0.971	0.96	0.985	0.985
iDDPM	0.973	0.966	0.976	0.979
PNDM	0.971	0.966	0.985	0.982
ADM_imag	0.655	0.616	0.733	0.77
SD	0.581	0.432	0.805	0.92
DM_mix	0.83	0.788	0.897	0.96
bigGAN	0.604	0.562	0.61	0.715
cycleGAN	0.508	0.507	0.492	0.711
starGAN	0.688	0.576	0.92	0.965
GAN_mix	0.6	0.548	0.674	0.776
Midjourney	0.594	0.532	0.501	0.920
DALLE2	0.615	0.755	0.811	0.952
Total_mix	0.744	0.698	0.813	0.881

Note: The highest values are highlighted in bold to emphasize the effectiveness of the adopted modules and the superiority of our proposed model.

similarity could facilitate better detection ACC compared to the more irregular patterns seen in bigGAN and cycleGAN images.

These performance variations highlight an area for improvement in the model's ability to generalize across images with substantial inherent differences. To enhance the model's robustness and broaden its applicability, future work should consider utilizing more diverse training datasets that ensure a more balanced representation of both GAN- and diffusion-generated images. Furthermore, exploring multimodal or hybrid training strategies could further augment the model's generalization capabilities across the diverse range of deepfake generation techniques.

3.7. Model Robustness Experiment Results and Analysis.

To further evaluate the robustness of the proposed model under adversarial conditions, we conduct a set of experiments to assess the model's resilience to common distortions and manipulations, including Gaussian blur and JPEG compression. Specifically, Gaussian blur is applied using a kernel with standard deviations of 0.5, while JPEG compression is tested with quantization factors of 80. In this experiment, we use the ProGAN training set, with the same mixed testing set as in Section 3.2, to evaluate different fusion strategies under Gaussian blur and JPEG compression. The performance metrics are summarized in Table 7.

The results indicate that Gaussian blur has a relatively limited impact on detection ACC, with a gradual decline observed as the blur intensity increases. The model, particularly when employing the AFF mechanisms, demonstrates the capability to maintain high detection ACC even under moderate blurring, highlighting its robustness against this type of distortion. On the other hand, JPEG compression shows a more pronounced effect on model performance, especially as the compression level becomes more severe.

These findings suggest that JPEG compression may have a considerable impact on frequency-domain features, potentially contributing to the observed performance

TABLE 7: Accuracy and F1-scores of models using different fusion strategies under Gaussian blur and JPEG compression, trained on the ProGAN dataset.

Fusing strategies Metrics	Concat		AFF1		AFF2	
	ACC	F1	ACC	F1	ACC	F1
None	0.799	0.779	0.8	0.777	0.815	0.795
Blur	0.785	0.776	0.786	0.747	0.798	0.77
Compression	0.711	0.685	0.703	0.692	0.758	0.708

degradation. However, even under these challenging conditions, the model demonstrates relatively good robustness, indicating its potential practical applicability in real-world scenarios where such common manipulations are often encountered. Additionally, the results of this experiment highlight the contribution of the AFF mechanism to the model's robustness. By incorporating the advanced fusion strategy AFF2, the model demonstrates improved resilience, making it better equipped to handle adversarial conditions and solidifying its potential as an effective solution for deepfake detection in diverse environments.

4. Conclusion

In this study, we propose a deepfake detection model that demonstrates superior detection ACC and generalization capability, effectively addressing the limitations of existing methods. The experimental results confirm that our model surpasses baseline methods in both detection ACC and generalization across various GANs and DMs, showcasing its robustness in identifying deepfake images from diverse sources. A key highlight of our model is its ability to generalize across different generative models: even when trained on the relatively dated ADM dataset, the model performs strongly on images generated by more recent DM Stable Diffusion v2 (SDv2) [30], as well as images generated by more advanced models like Midjourney [34] and DALLE2 [35], underscoring its robust generalization capability.

Such strong performance is largely attributable to the structure of our model. By leveraging multiscale frequency-domain features extracted by Gaussian, Gabor, and wavelet filters, and integrating them with multiscale spatial-domain features from a FPN, our model effectively captures complementary information across multiple scales and domains. The incorporation of the AFF mechanism further enhances the model's performance, as confirmed through ablation experiments.

In conclusion, the proposed model excels in detection ACC and generalization across diverse deepfake sources, positioning it as a robust solution for tackling the growing challenges in deepfake detection.

5. Discussion

While the proposed model demonstrates strong detection ACC and generalization capabilities across GAN- and diffusion-generated images, certain limitations remain. The model predominantly relies on frequency-domain features, with the FPN being utilized for spatial feature modeling. However, the spatial features captured by the FPN may not

fully reflect the intrinsic spatial characteristics of the images. Future work will aim to incorporate more explicit spatial-domain features, such as those derived from advanced denoising algorithms, edge-detection models, or texture analysis techniques. Integrating these explicit spatial features is expected to result in a more comprehensive and balanced feature extraction process, optimizing the overall architecture and enhancing the model's ability to detect diverse image manipulations.

Additionally, despite the notable generalization capabilities of the current model, the rapid evolution of generative models, particularly the increasing prevalence of fine-tuned variants, raises concerns about its adaptability to newly emerging and previously unseen content. Addressing this challenge will require further investigation into adaptive learning strategies that can accommodate the fast-paced advancements in generative models. Techniques such as continual learning or incorporating a wider variety of generative models during training may help maintain the model's effectiveness. Furthermore, as AIGC technologies continue to evolve, ensuring the traceability of manipulated content is becoming increasingly critical for maintaining transparency and security. Future research will extend toward improving detection capabilities for unknown generative models, focusing not only on adaptability but also on AIGC traceability to ensure the authenticity and security of generated content across diverse sectors.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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